

MID

$$1.a) \text{ Total time} = 62860800 = \frac{X}{8} \left(200 + 400 + \frac{8/1024}{96 \times 10^{-9}} \right)$$

$$\Rightarrow X = 6144 \text{ Bytes}$$

b) Burst Mode:

$$\text{Time} = 200 + 400 + \frac{6144/3}{1024} \cdot \frac{8/1024}{96 \times 10^{-9}} = 20833933.33 \text{ ns}$$

CS Mode:

$$\text{Time} = \frac{6144 \times 2/3}{8} \left(200 + 400 + \frac{8/1024}{96 \times 10^{-9}} \right)$$

$$= 41,973,866.67 \text{ ns}$$

$$\text{Total time} = 62,807,800 \text{ ns}$$

$$2.a) 76DA + 3867 = AF41 H$$

$$SF \rightarrow 1$$

$$PF \rightarrow 1$$

$$AF \rightarrow 1$$

$$CF \rightarrow 0$$

$$OF \rightarrow 1$$

- b) — Processor can/not execute directly from them
— Faster/Slower

3. a) We know,

$$S.A. = \text{Segment} * 10H$$

$$E.A. = \text{Segment} * 10H + FFFF$$

Now, for CS:

$$S.A. = 25A1 * 10H = 25A10 H$$

$$E.A. = 25A10 + FFFF = 35A0F H$$

Now, for DS:

$$S.A. = 4132 * 10H = 41320 H$$

$$E.A. = 41320 + FFFF = 5131F H$$

Now, for SS:

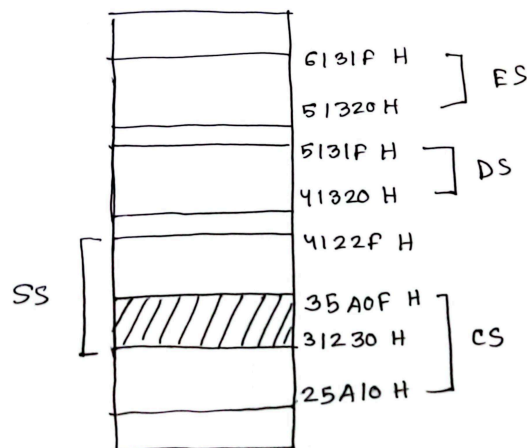
$$S.A. = 3123 * 10H = 31230 H$$

$$E.A. = 31230 + FFFF = 4122F H$$

Now, for ES:

$$S.A. = 5132 * 10H = 51320 H$$

$$E.A. = 51320 + FFFF = 6131F H$$



i) For overlapping segments of **CS and SS**,

$$\text{Overlapping slots} = (35A0F - 31230) + 1 = 47DF + 1 = 47E0 H$$

ii) From the above picture we can see that,

DS is placed **after SS and before ES** without being overlapped. So, it is possible.

b) Let,

The capacity of RAM Y is x MB.

So, the capacity of RAM X is $(\frac{2}{3})x$ MB.

$$\text{Total capacity} = x + (\frac{2}{3})x = (\frac{5}{3})x \text{ MB}$$

ATQ,

$$(\frac{5}{3})x = 15 \Rightarrow x = 9 \text{ MB So, RAM X} = (\frac{2}{3}) * 9 = 6 \text{ MB.}$$

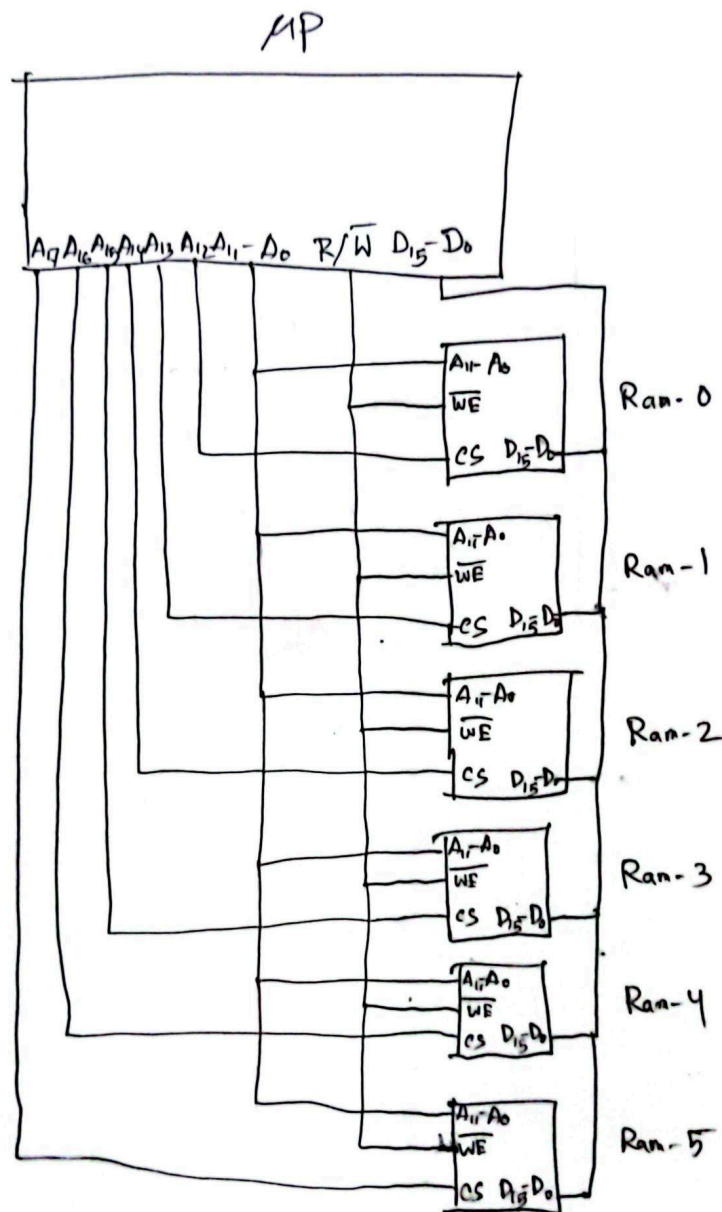
For RAM X,

$$2^{A.B.} \times D.B. = \text{Ram Size}$$

$$\Rightarrow 2^{22} \times D.B. = 6 * 2^{10} * 2^{10} * 8$$

$$\Rightarrow D.B. = 12 = 12 \text{ bit}$$

4. Ram capacity = $2^{12} * 16 \text{ bits} = 8 \text{ KB}$
 No, of RAMs required = $48 \text{ KB} / 8 \text{ KB} = 6$



CS Scanned with CamScanner

Address mapping

Ram - 03:

3rd address = 08002 H

5th address = 08004 H

7th address = 08006 H

Ram - 04:

3rd address = 10002 H

5th address = 10004 H

7th address = 10006 H

Ram - 05:

3rd address = 20002 H

5th address = 20004 H

7th address = 20006 H